

▣ MILAN 2026 – Official Website

National Cultural Festival of SRM Institute of Science and Technology

A modern, interactive Next.js web application showcasing MILAN 2026, featuring event management, team showcases, gallery exploration, and integrated payment systems.

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▣ Overview

MILAN 2026 is a comprehensive web platform designed for the annual cultural festival at SRMIST. The application provides:

- **Event Management:** Browse, register, and manage festival events
 - **Team Showcase:** Interactive 3D globe visualization of organizing teams
 - **Explore Gallery:** Draggable canvas with circular image placement
 - **Authentication:** Google OAuth integration via Supabase
 - **Payment Integration:** Konfuhub for event registrations and passes
 - **Responsive Design:** Optimized for desktop, tablet, and mobile devices
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▣ Features

▣ Home Page

- Hero section with custom animations
- Introduction to MILAN festival
- Poster generator for social sharing
- Directors and managers showcase
- Timeline of MILAN history
- Guest showcase section

▣ Events

- Browse all festival events
- Event registration system
- Event details with rulebook downloads
- Category-based filtering
- Responsive card-based layout

▣ Teams

- Interactive 3D globe with team markers
- Team member profiles with roles
- Smooth animations and transitions
- Responsive team listing
- Touch-optimized for mobile

▣ Explore Gallery

- **Circular Placement Algorithm:** Dynamic concentric ring layout
- **Draggable Canvas:** Pan and zoom functionality
- **Full-size Images:** Display at actual dimensions
- **Infinite Scroll:** Smooth exploration experience
- **Hashtag System:** Trending tags and filtering
- **Upload Feature:** Community-driven content

▣ Passes & Payments

- Digital pass generation
- QR code integration
- Konfuhub payment gateway
- Payment verification system
- Pass download functionality

▣ Authentication

- Google OAuth 2.0
- Supabase authentication backend
- Protected routes
- User session management

▣ Responsive Design

- Mobile-first approach
- Optimized navbar and footer
- Touch-friendly interactions
- Adaptive layouts

▣ Tech Stack

Frontend

- **Framework:** [Next.js 16](#) (App Router)
- **React:** 19.2.0
- **TypeScript:** ^5
- **Styling:** Tailwind CSS 4
- **Animations:**
 - Framer Motion (^12.23.24)
 - GSAP (^3.13.0)
 - Lenis (smooth scroll)

3D Graphics

- **Three.js** (^0.182.0) - Globe visualization

UI Components

- **Radix UI** - Accessible component primitives
- **shadcn/ui** - Component library
- **Lucide React** - Icon set

Backend & Database

- **Supabase** - Authentication, Database, Storage
- **PostgreSQL** - Database (via Supabase)

Payment

- **Konfuhub** - Payment gateway integration

Additional Tools

- **Nodemailer** - Email notifications
 - **Puppeteer** - PDF/poster generation
 - **QR Code Generator** - Pass validation
 - **JWT** - Token management
 - **UUID** - Unique identifiers
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▣ Getting Started

Prerequisites

- **Node.js** 20+ and npm/yarn
- **Supabase Account** ([Create one](#))
- **Google Cloud Project** (for OAuth)
- **Konfuhub Account** (for payments)

Installation

1. Clone the repository

```
git clone <repository-url>
cd milanDSA-frontend
```

2. Install dependencies

```
npm install
```

3. Set up environment variables

Create `.env.local` in the root directory:

```
# Supabase
NEXT_PUBLIC_SUPABASE_URL=your_supabase_url
NEXT_PUBLIC_SUPABASE_ANON_KEY=your_supabase_anon_key

# Email Settings (Nodemailer)
EMAIL_USER=your_email@example.com
EMAIL_PASS=your_email_app_password
```

```
# Base URL
NEXT_PUBLIC_BASE_URL=http://localhost:3000

# Konfuhub (optional, for payments)
KONFUHUB_KEY_ID=your_konfuhub_key_id
KONFUHUB_KEY_SECRET=your_konfuhub_key_secret
```

4. Configure Supabase

See [GOOGLE_AUTH_SETUP.md](#) for detailed Google OAuth setup.

5. Run the development server

```
npm run dev
```

6. Open your browser

Navigate to <http://localhost:3000>

📁 Project Structure

```
milanDSA-frontend/
├── app/                                # Next.js App Router
│   ├── api/                            # API routes
│   │   ├── auth/                      # Authentication endpoints
│   │   ├── explore/                  # Explore gallery API
│   │   ├── passes/                  # Digital passes API
│   │   ├── check-payment/           # Payment verification
│   │   └── webhooks/                # Payment webhooks
│   ├── checkout/                    # Payment checkout page
│   ├── event-register/              # Event registration
│   ├── events/                      # Events listing
│   ├── explore/                     # Gallery page
│   ├── gallery/                     # Photo gallery
│   ├── login/                       # Login page
│   ├── passes/                      # Digital passes
│   ├── sponsors/                    # Sponsors page
│   ├── team/                        # Team showcase
│   ├── layout.tsx                   # Root layout
│   ├── page.tsx                     # Home page
│   └── globals.css                  # Global styles
├── components/                       # React components
│   ├── ui/                          # shadcn/ui components
│   ├── event/                       # Event-related components
│   ├── explore/                     # Explore gallery components
│   ├── intro/                       # Intro animation
│   ├── sponsors/                    # Sponsor components
│   └── teams/                       # Team components
```

```

├── navbar.tsx          # Navigation bar
├── footer.tsx          # Footer
├── hero.tsx            # Hero section
├── draggable-canvas.tsx # Canvas for explore page
├── team-sphere.tsx     # 3D globe component
├── ...                 # Other components
├── lib/                # Utility functions
│   ├── database.types.ts # Supabase types
│   ├── circular-placement.ts # Image placement algorithm
│   └── ...              # Other utilities
├── context/            # React Context
│   └── auth-context.tsx # Authentication context
├── types/              # TypeScript definitions
├── public/             # Static assets
│   ├── Rulebooks/     # Event rulebooks
│   └── ...              # Images, fonts, etc.
├── supabase/           # Supabase configuration
│   └── migrations/    # Database migrations
├── scripts/            # Utility scripts
├── .env.local          # Environment variables (create this)
├── next.config.ts       # Next.js configuration
├── tailwind.config.ts  # Tailwind CSS config
├── tsconfig.json        # TypeScript config
├── package.json         # Dependencies
├── GOOGLE_AUTH_SETUP.md # Google OAuth guide
└── EXPLORE_NEW_IMPLEMENTATION.md # Explore feature docs

```

✳ Configuration

Next.js Configuration

The `next.config.ts` includes:

- Image domain whitelisting for Supabase storage
- Experimental features for performance optimization

Tailwind CSS

Custom configuration with:

- Custom fonts: `StampPress` , `Foglihten`
- Extended color palette
- Custom animations
- Responsive breakpoints

TypeScript

Strict mode enabled with path aliases:

- `@/` maps to root directory for clean imports
-

▣ Key Features Documentation

1. Explore Gallery - Circular Placement

See: [EXPLORE_NEW_IMPLEMENTATION.md](#)

The Explore page features a unique circular placement algorithm:

- **Dynamic Layout:** Images arranged in concentric rings
- **Full-size Display:** Uses actual image dimensions
- **Interactive Canvas:**
 - Pan: Click & drag / Touch & swipe
 - Zoom: Mouse wheel (30%-300%)
 - Reset: Button to return to center
- **Alternating Direction:** Clockwise → Counter-clockwise rings
- **Overlap Prevention:** Intelligent spacing algorithm
- **Random Center:** Different layout on each visit

Technologies:

- Custom placement algorithm
- GPU-accelerated transforms
- Next.js Image optimization
- Framer Motion animations

2. Team Sphere - 3D Globe

Interactive globe showcasing team members:

- **Three.js Integration:** Realistic 3D globe rendering
- **Interactive Markers:** Clickable team member locations
- **Smooth Rotations:** GSAP-powered animations
- **Touch Support:** Mobile-optimized interactions
- **Team Details:** Pop-up cards with member information

3. Authentication System

See: [GOOGLE_AUTH_SETUP.md](#)

Google OAuth integration:

1. **Google Cloud Setup:** OAuth 2.0 client configuration
2. **Supabase Provider:** Enable Google authentication
3. **Redirect URLs:** Configure allowed callbacks
4. **Session Management:** Persistent user sessions
5. **Protected Routes:** Route guards for authenticated pages

4. Payment Integration

Konfuhub payment flow:

1. User selects event/pass
2. Checkout page with payment details
3. Konfuhub modal integration
4. Payment verification webhook
5. Pass generation with QR code
6. Email notification

5. Event Registration

Complete registration workflow:

- Browse events with filters
- View event details and rulebooks
- Registration form with validation
- Payment processing
- Confirmation email
- Registration details page

▣ API Routes

Authentication

- `POST /api/auth/google` - Google OAuth callback
- `GET /api/auth/session` - Get current session
- `POST /api/auth/logout` - End session

Explore Gallery

- `GET /api/explore/posts` - Fetch all gallery posts
- `POST /api/explore/upload` - Upload new image
- `DELETE /api/explore/delete` - Remove post

Passes

- `POST /api/passes/generate` - Create digital pass
- `GET /api/passes/[id]` - Retrieve pass details
- `POST /api/passes/validate` - Validate QR code

Payments

- `POST /api/check-payment` - Verify payment status
- `POST /api/webhooks/konfuhub` - Payment webhook

▣ Deployment

Docker Deployment

Using the included Dockerfile:

```
# Build image
docker build -t milan-frontend .

# Run container
docker run -p 3000:3000 milan-frontend
```

Or use Docker Compose:

```
docker-compose up -d
```

Vercel Deployment

1. Push code to GitHub
2. Import project in Vercel
3. Add environment variables
4. Deploy

Environment Variables for Production

Ensure all environment variables are set:

```
NEXT_PUBLIC_SUPABASE_URL=  
NEXT_PUBLIC_SUPABASE_ANON_KEY=  
EMAIL_USER=  
EMAIL_PASS=  
NEXT_PUBLIC_BASE_URL=https://your-domain.com  
KONFUHUB_KEY_ID=  
KONFUHUB_KEY_SECRET=
```

📱 Mobile Optimization

The application is fully responsive with specific optimizations:

- **Navbar:** Hamburger menu with icon
- **Footer:** Condensed layout for small screens
- **Teams Page:** Touch-optimized scrolling
- **Explore Canvas:** Swipe gestures
- **Forms:** Touch-friendly inputs
- **Images:** Lazy loading and optimization

🔧 Development Scripts

```
# Development server  
npm run dev  
  
# Production build  
npm run build  
  
# Start production server  
npm run start  
  
# Lint code  
npm run lint
```



```
# Type check
npx tsc --noEmit
```

▣ Contributing

Code Style

- Use TypeScript for all new files
- Follow existing component patterns
- Use Tailwind CSS for styling
- Add comments for complex logic
- Keep components small and focused

Commit Guidelines

- Use descriptive commit messages
- Reference issue numbers
- Keep commits atomic

Pull Requests

1. Fork the repository
 2. Create feature branch (`git checkout -b feature/amazing-feature`)
 3. Commit changes (`git commit -m 'Add amazing feature'`)
 4. Push to branch (`git push origin feature/amazing-feature`)
 5. Open Pull Request
-

▣ Troubleshooting

Build Errors

Issue: "Module not found"

- **Solution:** Check import paths, ensure `@/` alias works
- Run `npm install` to ensure dependencies are installed

Issue: TypeScript errors

- **Solution:** Run `npm run build` to see all errors
- Check `tsconfig.json` paths configuration

Authentication Issues

Issue: "Provider is not enabled"

- **Solution:** Enable Google OAuth in Supabase Dashboard
- See [GOOGLE_AUTH_SETUP.md](#)

Issue: "redirect_uri_mismatch"

- **Solution:** Verify callback URL in Google Cloud matches Supabase
- Check allowed redirect URLs in Supabase settings

Performance Issues

Issue: Slow page loads

- **Solution:**
 - Optimize images (compress before upload)
 - Enable Next.js Image optimization
 - Check network tab for large assets

Issue: Canvas lag on Explore page

- **Solution:**
 - Reduce number of images
 - Disable animations temporarily
 - Check browser console for errors

Database Issues

Issue: Supabase connection fails

- **Solution:**
 - Verify environment variables
 - Check Supabase project status
 - Ensure correct URL and anon key

Issue: Migration errors

- **Solution:** Check `supabase/migrations/` folder
- Run migrations manually via Supabase Dashboard

▣ Additional Documentation

- [GOOGLE_AUTH_SETUP.md](#) - Complete Google OAuth setup guide
- [EXPLORE_NEW_IMPLEMENTATION.md](#) - Explore gallery technical details

▣ Support

For issues or questions:

- **Technical Team:** techteam.sa@srmist.edu.in
- **Documentation:** Check the guides in the repository
- **Issues:** Open a GitHub issue

▣ License

This project is proprietary and confidential. All rights reserved by SRM Institute of Science and Technology - Student Activities.

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Built with ♥ for MILAN 2026

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